

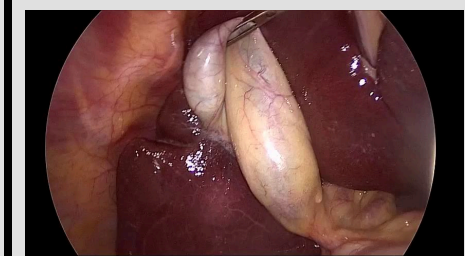
Input

Natural Language Query

"During the calot-triangle-dissection phase, the grasper is used to retract the omentum."

Surgical Video

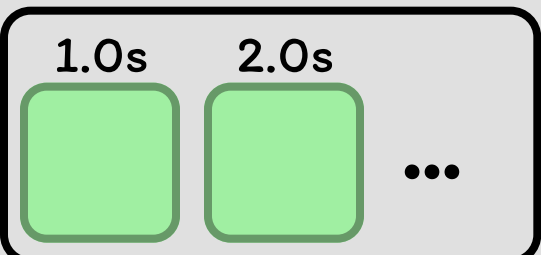
Chunk 1



Chunk t

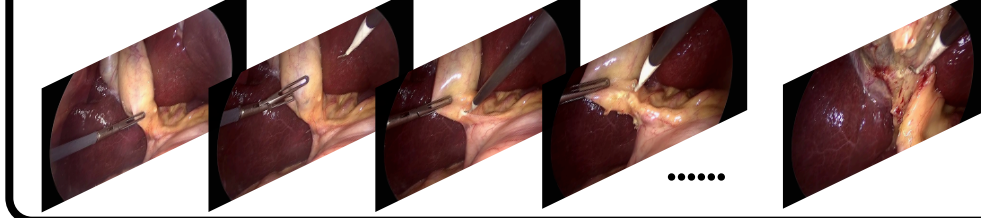


Timestamps

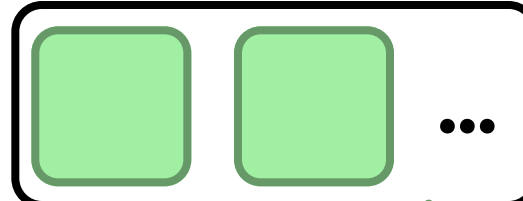


Surgical Video Frames

Chunk t

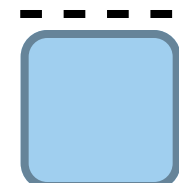
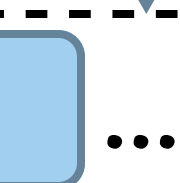
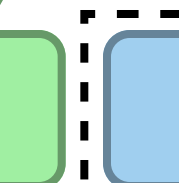
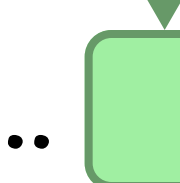
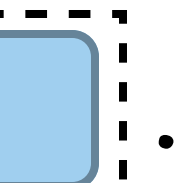
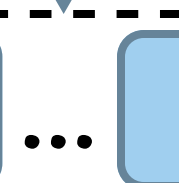
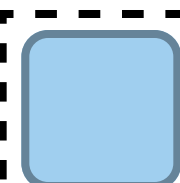
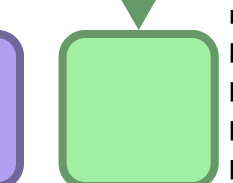
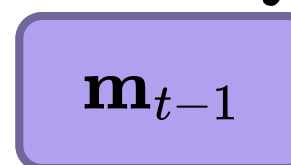


Timestamps

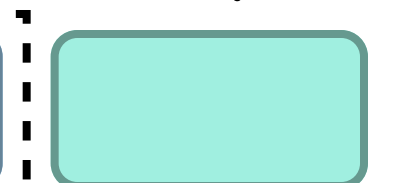


Vision Encoder & Projector

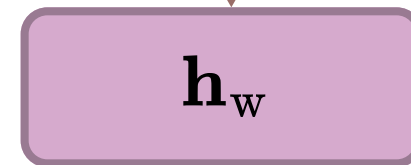
Read
Memory



Task
Description



Large Language Model



f_{slot1}

f_{slot2}

$f_{\text{c-inst}}$

$f_{\text{b-inst}}$

$f_{\text{b-tri}}$

$\mathcal{L}_{\text{mem}}$

Macro (phase)

Meso (Inst)

Micro (Triplet)

$\mathbf{m}_t$

$\mathcal{L}_{\text{CE}}$

<answer>

70.0s-75.0s

91.0s - 96.0s

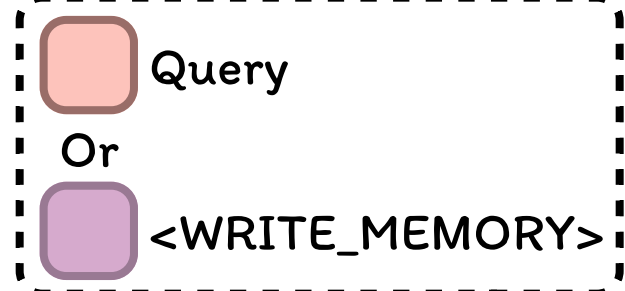
</answer>

Answer
Prediction

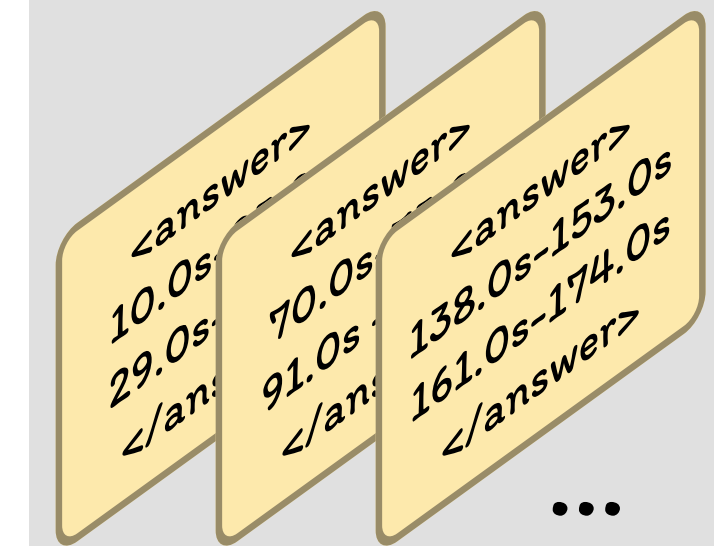
Hidden States of
<WRITE_MEMORY>

MLP Heads

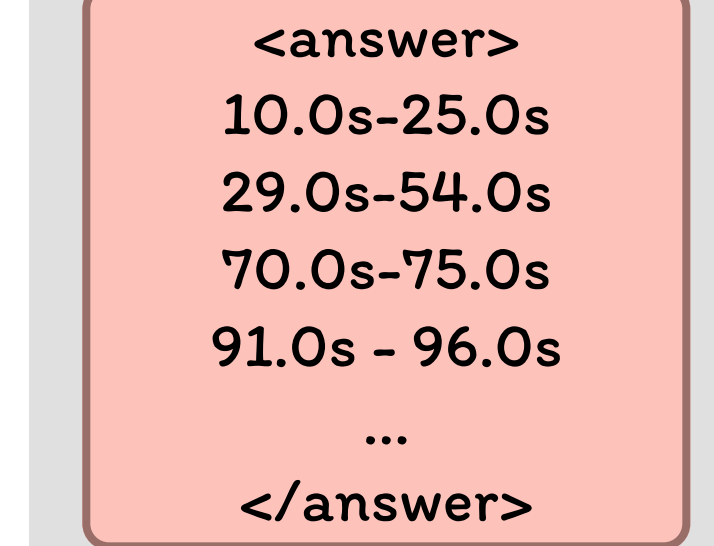
Write Memory



Output



Per-Chunk Answers



Final Answer